

IO-X: Detecting and Attributing Content-Duplicating Influence Operations on X (Twitter)

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Abstract

Social platforms' data-access restrictions hinder independent study of influence operations (IOs). We present a formal model of IO undetectability, frame it as a security game and show that content duplication induces detectable gaps. We introduce IO-X, a system for detecting multimodal, content-duplicating IOs (CD-IOs). IO-X links near-duplicate text and images via embedding-based multi-graph analysis with sentiment filtering. This uncovers meme-style campaigns missed by text-only methods. It attributes campaigns using ForeignScope, that combines language detection, belief propagation, and timezone inference to flag foreign-linked accounts, and PAN, which integrates profile, content, network, and LLM-derived features for political alignment. Applied to fact-checker-seeded data from the 2024 U.S. and Indian elections, IO-X uncovered 27k+ CD-IOs in the U.S. and 37k+ in India, including 30k+ campaigns duplicating false information, specious sources, or hate speech, and revealing strong partisan asymmetries.

CCS Concepts

• **Information systems** → **Social networks**; *Web mining*; • **Theory of computation** → Security protocols.

Keywords

Influence operations, coordinated inauthentic behavior, information integrity, social networks, foreign-linked activity

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1 Introduction

Social media platforms enable unprecedented reach, allowing individuals and organizations to rapidly shape public discourse. However, features that make platforms open and participatory, e.g., low barriers to account creation, algorithmic amplification, and limited identity verification, also enable inauthentic influence operations (IOs). These operations manipulate information flows by spreading

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The Biden administration stole **\$1 billion from FEMA to use it for illegal migrants**. And FEMA is now busted (Source: PolitiFact; X/Twitter effect: 22 duplicates, 19.42 M views)

There are **20 million votes MISSING for Kamala**. No way there were 20 million fewer votes than 2020 with a massive 25 million surge in voter registration and record turnout. **NO WAY**. (Source: PolitiFact; X/Twitter effect: 31 duplicates, 3.3 M views)

Figure 1: Sample tweets from campaigns to duplicate information fact-checked to be false by PolitiFact [4].

false claims, hate speech, and polarizing narratives that undermine trust in online spaces and democratic debate [29, 42, 44, 55, 58, 59]. Detecting IOs and enabling independent, transparent oversight remains challenging under adversarial conditions.

Major platforms invest substantial resources in detecting and removing IOs [8, 10, 20] and periodically release datasets to promote transparency [7–9, 51]. However, these releases are selective, retrospective, and opaque about detection methods. This limits independent assessment and risks false positives. In response, researchers have pursued external approaches, including analyzing IO strategies [29, 32, 53], building detectors from metadata and content [37, 42, 58], clustering or matching content [24, 47], and attributing accounts via political alignment [26, 44]. Most such methods, however, focus on text alone, overlooking near-duplicate images and memes central to modern IOs.

In addition, two limitations hinder responsible IO research. First, most detection and attribution efforts rely on platform-released datasets that provide only static, retrospective views of operations and fail to capture evolving tactics. Second, independently scraped data is increasingly constrained by platform restrictions. This chills efforts to verify platform claims or track IOs in real time. These constraints raise fundamental questions about how IOs can be studied responsibly. We therefore ask:

- **(RQ1)**. Can we design a transparent and reproducible system that detects and analyzes influence operations using only publicly accessible platform data?
- **(RQ2)**. Does near-duplicate detection for both text and images uncover operations missed by text-only approaches?
- **(RQ3)**. What structural and behavioral patterns of influence operations emerge from transparent large-scale detection?

To address these questions, we develop a formal model of influence operations and their strategic objective: maximizing narrative impact while remaining undetectable to adversaries (platforms, researchers). We formalize undetectability via a security game-inspired abstraction, that captures when coordinated promotion cannot be distinguished from background behavior. Building on

this abstraction, we derive an analytical framework that characterizes when IO detectors achieve non-negligible advantage, and identify conditions under which transparent, reproducible oversight of influence operations is possible despite platform restrictions.

Building on platforms' treatment of large-scale duplication as a marker of inauthenticity [40, 54, 60], we define *content-duplicating influence operations* (CD-IOs) as operations driven by groups of accounts disseminating identical or near-identical content to promote shared narratives. Duplication is both their central strategy and the signal that breaks undetectability: coordinated accounts create a measurable duplication gap distinguishing them from organic activity (Figure 1). Making this signal explicit enables transparent external oversight.

We introduce the Influence Operations eXtractor (**IO-X**), a system for detecting content-duplicating influence operations (CD-IOs) on X/Twitter using only publicly accessible data. IO-X seeds detection from fact-checker reports, leveraging the insight that accounts repeating verified false claims are more likely to engage in coordinated campaigns. It links near-duplicate text and images, an advance over prior text-only methods [36, 42, 45, 58, 59], and incorporates coordination and narrative-extraction components to provide interpretable views of detected campaigns.

IO-X advances the state of the art in three ways. First, it integrates near-duplicate text and image detection, exposing both meme-based and textual campaigns. Second, its ForeignScope module combines language detection, graph inference, and posting-time analysis to reveal foreign-linked activity beyond superficial cues. Third, its PAN module fuses content, profile, and network features to infer political alignment even when accounts mask partisanship. These independent signals raise the cost of evasion and enable transparent detection and attribution under platform restrictions.

To evaluate how responsible, reproducible analysis can be carried out under real-world access constraints, we investigate IO-X on datasets seeded from PolitiFact (U.S.) and AltNews (India) spanning the 2024 election periods. For RQ1 and RQ2, IO-X substantially outperforms prior duplication methods [45, 59], detecting over 30× more campaigns in U.S. data and over 40× more in Indian data, with validation confirming true duplicates. Its inclusion of near-duplicate images further expands account graphs, and yields up to 4.8× more accounts and 32× more ties than prior work. PAN further improves political orientation classification by 18-22 F1 points.

For RQ3, IO-X provides structural and behavioral insights: it (1) identifies over 60k IOs across U.S. and Indian groups and 30k+ campaigns spreading false claims, untrustworthy links, and hateful content (many drawing millions of views), (2) reveals cross-border participation in U.S. but not Indian groups, and (3) exposes partisan asymmetries (e.g., 3× more Republican than Democratic IOs; BJP dominance over INC). Thus, IO-X enables independent IO detection and supports accountability and transparency on today's web.

The source code of IO-X is available for download at <https://github.com/shafin191/IOX>. The anonymized data collected in this research is available at <https://github.com/shafin191/IO-X-Datasets/>. Since X no longer has contact channels except for law enforcement/government agencies, we have reported novel findings to PolitiFact [4], AltNews [1] and the Center for Countering Digital Hate (CCDH) [2].

2 Related Work

Coordinated Behavior and Inauthenticity. Coordinated behavior (CB) on social media involves synchronized actions by multiple accounts to amplify narratives. While coordination can be organic [44], inauthentic manipulation is often characterized by rapid resharing and content duplication [45, 59]. Prior work detects CB via temporal synchronization [45, 56, 58] and content similarity [12, 26, 37, 42, 44, 45, 59]. Vargas et al. [58] further distinguish authentic signals (hashtags, URLs) from inauthentic ones (duplication) and integrate these features into classifiers [36, 42].

We focus on content duplication, a known indicator of inauthenticity and a violation of X/Twitter policy [60]. Prior work detects duplication via pattern recognition [49] over short sliding windows (e.g., 10 tweets) [45, 59]. In contrast, IO-X analyzes entire timelines and uses FAISS search [18] to detect near-duplicate text and images.

Influence Operations. Prior work on influence operations (IOs) studies coordination strategies and detection methods [29, 32, 53]. Luceri et al. [37] identify coordinated actors using similarity graphs and temporal community detection, while Minici et al. [42] fuse content embeddings with behavioral similarity networks to detect Twitter IO users. Other work focuses on IO attribution via political alignment [26, 44], yielding non-negligible advantage in the IO-UND game. IO-X builds on these efforts by targeting long-term, misinformation-driven IOs attributed to organizational groups. Unlike prior methods limited to short time windows [45, 59], IO-X detects duplication across months or years and supports real-time analysis, rather than relying on static platform-released datasets [9, 51].

Prior work also tracks misinformation narratives. Hanley et al. [24] cluster content from unreliable news sites to assist fact-checkers. Paudel et al.'s Lambretta system [47] identify semantically similar claims in tweets. IO-X complements these approaches by attributing narratives to organizational actors, using BERTopic [23] to extract coherent narratives from long-term duplication campaigns.

Political Orientation Inference. Early work on political-orientation inference focused on politicians and their supporters [15, 21, 22, 46]. Later studies refined content-level polarity inference [44]. However, Cohen and Ruths [17] showed that such models generalize poorly beyond highly political users. This is exacerbated in IOs where operators mask partisanship. PAN addresses these gaps by integrating content, profile, and network signals, using a transformer-based model to assess post polarity and introducing a "mix" category for accounts aligned with multiple political groups.

Account Localization. Prior work infers user location from tweet content [16, 19], account metadata, or geographic information systems [27]; ensemble methods combine content and sharing behavior to improve accuracy [38]. ForeignScope avoids reliance on unreliable self-reported location or language labels to instead identify foreign-linked accounts by combining multiple language detection models with loopy belief propagation and posting-time analysis.

3 Model

3.1 Modeling the Social Platform

We consider the X/Twitter social platform, where users post content, follow others, and interact through likes, retweets, and views. The platform also exposes engagement counts (e.g., retweets, likes, impressions) as measures of reach.

Access to platform data is restricted. Twitter previously offered free academic API access, but after the 2022 acquisition X replaced the API with paid tiers only that allow far fewer requests. Researchers must now rely on expensive subscriptions or third-party scrapers (e.g., Twikit [6]). Both are subject to rate limits far more restrictive than those of the legacy Twitter API. Thus, comprehensive collection of user timelines is now infeasible.

These restrictions make comprehensive collection of user timelines infeasible and significantly limit what researchers can observe.

3.2 Modeling Influence Operations

Let $\mathcal{U}$ be the set of all accounts on the social platform, and $\mathcal{O}$ a set of organizational entities of interest (e.g., political or non-political, government bodies, interest groups), $n = |\mathcal{O}|$.

Influence Operations are coordinated efforts by groups of accounts to support an organization, advance a topic, and promote a narrative using specific strategies. Topics are neutral subjects of posts, while narratives convey structured messages, emotions, or morals. In doing so they seek to shape public perception, opinion, or behavior to achieve strategic goals [20, 55, 57].

Influence Operations Goals. IOs typically pursue two objectives. First, to maximize narrative impact by encouraging widespread engagement, ideally leading to adoption where narratives detach from their origin and spread organically. Second, to remain undetected by platforms, researchers, and rivals, thus avoid bans, removals, or countermeasures. Undetectability is thus crucial for sustained influence, and we formalize it next.

Influence Operations Undetectability (IO-UND). The IO-UND security game takes place between a challenger C (the IO operator) and an adversary $\mathcal{A}$ (e.g., a platform or researcher). At the start, C controls a group G of accounts and secretly chooses a random bit $b \in \{0, 1\}$. Across r rounds, $\mathcal{A}$ nominates topics and narratives and monitors the platform. If $b = 1$, C coordinates G to promote the nominated narratives; if $b = 0$, C has G behave arbitrarily without targeting them. At the end of the interaction, $\mathcal{A}$ outputs b' , its guess for the hidden bit. The advantage of $\mathcal{A}$ in this game is $\text{Adv}(\mathcal{A}) = |\Pr[b' = b] - \frac{1}{2}|$: IOs achieve *undetectability* if no efficient adversary can obtain non-negligible advantage, i.e., coordinated promotion cannot be distinguished from background behavior.

4 Analytical Framework

We introduce an influence operations-detection framework that focuses on the subclass of content-duplicating IOs (CD-IOs), defined below. This subclass is detectable because coordinated accounts leave measurable duplication signals. The framework's practical value lies in offering theoretical guarantees on when independent researchers can expose coordinated IOs. This enables reproducible oversight despite constrained data access.

Duplication Campaigns. A duplication campaign is a tuple $DC = \langle C, G, TL \rangle$, where C is duplicated content, $G \subset \mathcal{U}$ the accounts posting it, and TL the chronological timeline of posts.

Content-Duplicating Influence Operations. A content-duplicating influence operation (CD-IO) consists of one or more duplication campaigns promoted by a group of accounts from $\mathcal{U}$ to advance a topic and narrative in support of an organization from $\mathcal{O}$.

IODetSys Definition. We define IODetSys = $(K, \mathcal{O}, \text{Mon}, \text{OpEx}, \text{OpAt})$ as a framework that, given knowledge K , detects CD-IOs (§ 3.2) and attributes them to organizations in $\mathcal{O}$. IODetSys requires three capabilities: (1) *Mon* to collect posts under restricted data access; (2) *OpEx* to detect CD-IOs; and (3) *OpAt* to attribute coordinated groups to organizations. The detector $\mathcal{A}$ uses it in the IO-UND game: if IODetSys returns a non-empty set of CD-IOs, $\mathcal{A}$ outputs $b' = 1$; otherwise, $b' = 0$.

Claim 1 (Capability necessity). *Mon, OpEx, and OpAt are jointly necessary for IODetSys to achieve non-negligible advantage in detecting and attributing CD-IOs.*

The proof is included in Appendix A. An implication of Claim 1 is that solutions based solely on published datasets [28, 34, 36, 37, 39, 42, 52, 58, 62] do not instantiate the full IODetSys framework, since they lack the *Mon* component. Monitoring-based approaches [41] address *Mon* and *OpEx* but not *OpAt*. This limits their ability to attribute detected IOs in the IO-UND game.

In the following we introduce the capabilities and conditions under which an influence-operations detector $\mathcal{A}$ using IODetSys can achieve non-negligible advantage in detecting and attributing CD-IOs in the IO-UND game.

Observation Parameters. Let p_+ be the probability that an IO account is returned by the platform's search when posting on an adversary-chosen topic and narrative during the IO-UND game. Let p_- be the probability that a background (non-IO) account is returned by the same search. These are the observable rates from the detector's perspective. We define the *duplication gap* as $\Delta \triangleq p_+ - p_-$, the excess likelihood that IO accounts appear on a nominated topic–narrative compared to background accounts.

Minimal Co-Activity Test. In the IO-UND game, let X be the size of the largest same-round co-activity set among collected accounts, i.e., the number of accounts observed to post on all r nominated topic–narrative pairs. This is the weakest cross-round co-ordination statistic: detectors in practice can use richer features (near-duplicates, time lags, partial overlaps). However, X suffices to capture the core phenomenon. Let an IODetSys detector with capabilities $(\text{Mon}, \text{OpEx}, \text{OpAt})$ and let u be the number of accounts returned by *Mon*. Let g be the number of accounts in G , controlled by the challenger C in the IO-UND game. IODetSys computes X and declares $b' = 1$ iff $X \geq s$, where s is a parameter threshold. We then show the following:

PROPOSITION 1. *If $\Delta > 0$ and the expected IO co-activity $gp_+^r = \Omega(\log(1/\epsilon))$ for some constant $\epsilon \in (0, 1)$, then the advantage $\text{Adv}(\mathcal{A})$ achieved by $\mathcal{A}$ using IODetSys is non-negligible.*

The proof is provided in an extended version. This framework abstracts platform mechanics (e.g., recommender dynamics, non-stationarity, feedback loops) and treats duplication as the primary signal. Other cues (e.g., sockpuppet reuse, financial flows) can be modeled by redefining X . Prop. 1 does not imply all IOs are detectable: if $p_+ \approx p_-$, the duplication gap Δ vanishes. Detection advantage grows with coverage but shrinks if fact-checking or keyword search miss posts. Lower thresholds improve recall but reduce precision, while more monitored rounds amplify coordination signals but at the cost of timeliness and robustness.

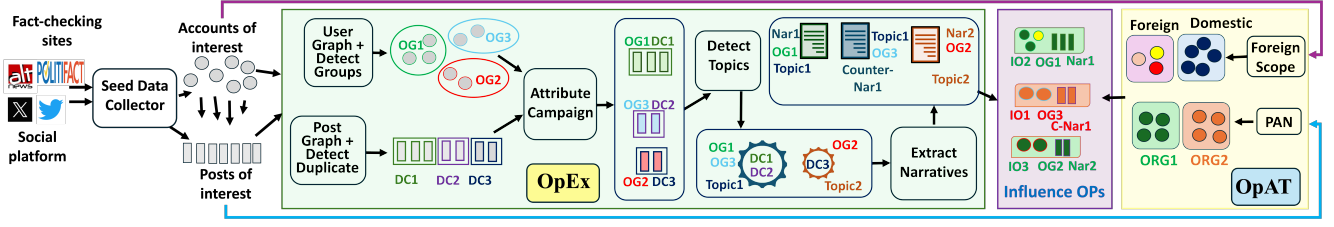


Figure 2: IO-X architecture. Given false information reported by a fact-checking site, IO-X identifies content-duplicating influence operations organized by the X/Twitter accounts.

5 IO-X

IO-X instantiates the IODetSys framework (§ 4) by implementing *Mon*, *OpEx*, and *OpAt* with concrete algorithms. By construction, IO-X inherits the guarantees of Proposition 1: when a duplication gap exists, IO-X achieves non-negligible advantage in the IO-UND game. That is, it can reliably distinguish coordinated IOs orchestrated by an operator from background activity.

5.1 Monitoring Under Platform Constraints

To address the data-access limits IO-X leverages the hypothesis that accounts spreading verified false content are more likely to engage in coordinated campaigns. For this, the *Mon* function seeds its search for candidate accounts for monitoring from fact-checking reports. For each report, it retrieves referenced tweets or searches for matching claims, collects the posting accounts, and inserts them into a priority queue. The queue favors active accounts (higher posting rates, more recent activity), ensuring frequent updates for the most relevant users while discarding inaccessible ones. Algorithm 1 in Appendix B details the monitoring process.

5.2 OpEx: Detecting IOs at Scale

The *OpEx* (operations extractor) function in IO-X detects influence operations by analyzing data from the *Mon* function. It identifies groups of accounts involved, the posts they have duplicated, and the topics and narratives they promoted. Using posts returned by *Mon* (§5.1), *OpEx* first finds content duplication campaigns (*CDC*), then detects account groups leveraging multiple campaigns to push a topic and narrative, see Figure 2.

Building Similarity Graphs. *OpEx* extends previous work by identifying identical or near-identical posts across both text and images. Image duplication is particularly challenging: posts may vary in dimensions, file size, format, or color channels (e.g., RGB vs. grayscale), and may even embed different overlaid text. To handle this, *OpEx* applies transformer-based encoders to generate embeddings for each post’s text and image content, then uses FAISS [18] to retrieve pairs whose cosine similarity exceeds thresholds T_{th} (text) or I_{th} (images).

From these candidate pairs, *OpEx* constructs three complementary graphs: (1) the post-similarity graph (PSG), linking posts by overall similarity; (2) the text-similarity graph (TSG), capturing near-duplicate text; and (3) the account-similarity graph (ASG), connecting accounts that share duplicated content. Text-similar pairs contribute edges to all three graphs. Image-similar pairs are considered only if the posts were not already linked by text, and

only when their accompanying text has consistent sentiment or one post lacks text. In such cases, *OpEx* adds an edge in PSG and increments the corresponding ASG weight. This design allows image similarity to strengthen duplication signals without conflating visually related but semantically contradictory posts.

Identify Organization Groups. *OpEx* leverages common duplication behaviors recorded for the accounts of interest in the account similarity graph ASG to identify the groups of accounts behind the identified duplication campaigns. More specifically, *OpEx* uses the Louvain community detection algorithm [11] to partition ASG into G_i , $i = 1..s$, a disjoint set of groups of accounts, such that the accounts in a group have duplicated more content together than with accounts in other groups.

Attribute Duplication Campaigns. The Louvain algorithm [11] extracts content duplication campaigns from the text-similarity graph (TSG). A resulting duplication campaign $DC = \langle C, A, TL \rangle$, where C is the duplicated content, A the participating accounts, and TL the timeline of posts, is said to be θ_c -controlled by a group G if $|A \cap G| > \theta_c |A|$, i.e., members of G produce at least a fraction θ_c of its posts. *OpEx* uses this criterion to assign campaigns to the operative groups identified earlier. We evaluate θ_c in §6.2.

Unlike prior work that clusters posts solely by text duplication, *OpEx* integrates text and image embeddings with sentiment filters, and uses a multi-graph structure (PSG, TSG, ASG) to both detect campaigns and attribute them to operative groups. This combination enables robust identification of narratives, which are then abstracted into influence operations.

Extract Promoted Topics and Narratives. *OpEx* derives topics and narratives from the campaigns promoted by each operative group. It first applies a topic modeling framework (e.g., BERTopic [23]) to cluster the content C of duplication campaigns into topics. For each topic, it records a *support cluster*: the set of campaigns containing that topic. It then associates each operative group with the subset of support campaigns it controls. For each topic-group pair, *OpEx* aggregates the posts from those campaigns, canonicalizes near-duplicates, orders the posts chronologically, and prompts an LLM to summarize the narrative they collectively advance. The resulting tuple $(G, \text{topic}, \text{narrative}, \text{support campaigns})$ is added to the set of detected influence operations. Attribution to specific organizations is deferred to *OpAt*.

Narratives as influence operations. Each narrative defines one influence operation, though a group may promote multiple narratives. Topics serve only as anchors: depending on which groups control the associated campaigns, a topic may give rise to different narratives. Three cases occur: (i) a topic controlled by a single group

yields a single narrative/operation; (ii) a topic controlled by multiple groups that support the same organization yields a shared narrative but distinct group-specific operations; and (iii) a topic controlled by groups aligned with different organizations is split, producing separate narratives and operations for each group.

5.3 OpAT: Transparent IO Attribution

Understanding the structure and intent of account groups conducting influence operations requires two tasks: detecting foreign-linked accounts and determining political alignment. The former uncovers inauthentic cross-national efforts and attempts to manipulate international audiences. The latter attributes groups to political factions, thus provides insights into strategic objectives and target audiences. We introduce ForeignScope and PAN, tools developed to perform these tasks.

ForeignScope flags foreign-operating accounts using two tweet-level signals: language and posting times. For language detection, it applies majority voting across three models: X/Twitter’s classifier, FastText [31], and a fine-tuned XLM-RoBERTa [35]. It then labels an account as writing in language L if it has posted at least one tweet in L . ForeignScope then applies Loopy Belief Propagation (LBP) [43] to expand from seed foreign-language accounts. Unlike supervised classifiers, LBP leverages the social graph, harder to manipulate than content features, to propagate labels. This enables detection of foreign-linked accounts that post only in the target language but remain socially tied to known foreign actors.

We formulate detection of accounts writing in language L as a network classification problem over the ASG graph $(\mathcal{U}, \mathcal{E}_w)$. The graph is modeled as a pairwise Markov Random Field (MRF) [61], where each node $u_i \in \mathcal{U}$ has a random variable $Y_i \in \mathcal{L} = \{L\text{-write}, \text{not-write}\}$ indicating whether u_i writes in L . By the Markov property, each label depends only on its neighbors. The joint distribution is the normalized product of node and edge potentials [33]:

$$\mathcal{P}(\mathbf{y}) = \frac{1}{Z} \prod_{Y_i \in \mathcal{U}} \phi_i(y_i) \prod_{(Y_i, Y_j, w_{ij}) \in \mathcal{E}_w} \psi_{ij}^w(y_i, y_j),$$

where $\mathbf{y}$ is a label assignment, w_{ij} the number of duplicates between u_i and u_j , and Z the normalization constant. Here $\phi : \mathcal{L} \rightarrow \mathbb{R}^+$ gives node priors and $\psi^w : \mathcal{L} \times \mathcal{L} \rightarrow \mathbb{R}^+$ the compatibility potentials.

The priors ϕ_i and potentials ψ_{ij} are adapted to X/Twitter user dynamics. Seed accounts receive priors $\phi_i = 1$, while all others are set to $1/2$. The compatibility potential $\psi_{ij}(y_i, y_j)$ models the likelihood that accounts u_i and u_j with labels y_i, y_j are neighbors linked with weight w_{ij} . For parameter β ,

$$\psi_{ij}(y_i, y_j) = \begin{cases} e^{\beta w_{ij}}, & y_i = y_j, \\ e^{-\beta w_{ij}}, & y_i \neq y_j. \end{cases}$$

Following the expansion step, ForeignScope infers an account’s geographic timezone from tweet and retweet timestamps, identifying inactivity windows of at least six hours paired with daytime activity aligned to a specific timezone. Overall, ForeignScope flags as foreign-linked those accounts that (1) either post multiple tweets in a foreign language or are connected via LBP to such accounts, and (2) exhibit activity patterns consistent with the daylight cycle of the corresponding regions.

Table 1: Collected datasets and corresponding duplication campaigns identified based on tweet text and image.

Source	Dataset Description					Only Text	Text+Image
	# Reports	# Tweets	# Users	# Text	# Image	# Campaigns	# Campaigns
US-ELECT24	2,207	1.8M	13.1K	1.3M	473K	65.1K	76.6K
IND23	433	6.88M	5.4k	5.2M	1.5M	187K	210.6K
IND-ELECT24	576	355K	4.7k	270K	146K	9.9K	12.5K

PAN: Political Alignment Notation. To attribute influence operations to responsible organizations, IO-X uses the Political Alignment Notation (PAN) component to classify an account $acc \in \mathcal{U}$ based on its public support for organizations in $\mathcal{O}$. PAN is the first classifier to combine profile, content, and network data, and to leverage large language models (LLMs) for fine-grained political annotation. Specifically, PAN applies three prompts to a modified DeBERTa model [13] to assess whether each post of the account is (1) political, and whether it (2) supports or (3) opposes any organization in $\mathcal{O}$. From these outputs, PAN derives support/oppose rates for each organization, together with apolitical proportions.

When LLMs lack sufficient context, such as for foreign organizations or non-English posts, PAN uses manually curated keyword lists for each organization in $\mathcal{O}$, (e.g., names, notable members) to extract features based on keyword matches in the account’s posts. PAN also builds a bag-of-words model from the latest 20 posts across accounts, selecting the 25 most frequent words and using their TF-IDF scores as features. Similar features are extracted from the account’s profile text. Additionally, PAN computes sentiment features, capturing the percentage of tweets expressing positive, negative, or neutral sentiment toward each organization in $\mathcal{O}$.

PAN trains a supervised classifier with $n + 2$ classes (one for each organization in $\mathcal{O}$, where $n = |\mathcal{O}|$, plus neutral and mixed) to predict account-level alignment. It then attributes groups identified by *OpEx*: an organization $O \in \mathcal{O}$ θ_o -controls a group G if at least a fraction θ_o of accounts in G are PAN-attributed to O ; otherwise, G is labeled mixed-support.

6 Evaluation

6.1 Data Collection

To evaluate whether IO-X can detect and analyze IOs transparently using only publicly accessible platform data (RQ1), we implemented the *Mon* function of IO-X in Python using Selenium [5]. *Mon* collects data starting with fact-checked misinformation from PolitiFact [4] and AltNews [1], as summarized in Table 1.

PolitiFact. Between July and November 2024, *Mon* collected 2,639 reports from PolitiFact [4]. Of these, 2,207 rated False or Pants on Fire constitute the **PolitiFact24** dataset. *Mon* identified 13,157 X accounts that duplicated content from PolitiFact24, producing 20,186 tweets. These accounts form the **PolitiX** dataset. *Mon* monitored these accounts around the U.S. elections (July–December 2024), producing the **US-ELECT24** dataset with 1.86M unique tweets.

AltNews. In April 2022, *Mon* collected 433 fact-check reports from AltNews [1] between 2020 and 2022. This forms the **AltNews20-22** dataset. Between 2022 and 2024, *Mon* collected an additional 576 reports, comprising the **AltNews24** dataset. *Mon* identified 5,493 X accounts that duplicated misinformation in AltNews20-22, producing the **AltX** dataset. Between February 13–15, 2023, *Mon* collected 6.88M tweets from the AltX accounts, from which

Table 2: *OpEx* content duplication detection significantly outperforms existing methods, in both U.S. and IND datasets.

	Method/Source	# Campaigns	# Tweets	Nodes	Edges
US-ELECT24	Pacheco et al. [45]	2,368	4,679	222	286
	Vishnuprasad et al. [59]	3,986	8,108	958	1,305
	<i>OpEx</i> (Only Text)	63,112	162,930	3,762	28,236
	<i>OpEx</i> (Text + Images)	74,598	208,987	4,568	41,930
IND23	Pacheco et al. [45]	4,517	9,442	345	437
	Vishnuprasad et al. [59]	19,190	39,772	1,301	7,021
	<i>OpEx</i> (Only Text)	174,936	598,781	1,724	15,345
	<i>OpEx</i> (Text + Images)	198,457	675,543	2,077	30,047

we curated **IND23**, a dataset of 4.33M Hindi and English tweets. *Mon* monitored AltX accounts during the Indian election period (April-October 2024), yielding **IND-ELECT24** with 355K tweets.

Ethical Considerations. Data collection was IRB-approved and declared exempt. We used only publicly accessible X/Twitter data (via Twitter Academic API until May 2023, then Twikit [6]). We anonymized user information, and excluded private content.

6.2 Operation Extraction (*OpEx*) Evaluation

Evaluating Duplication Detection At Scale (RQ2). To assess the impact of multimodal duplication detection (RQ2), we combined text and image similarity measures. Tweet text was preprocessed by removing links, mentions, and hashtags, and embeddings were generated using Sentence-BERT (paraphrase-multilingual-mpnet-base-v2) [50]. Images were resized to 224×224 pixels, converted to grayscale, and embeddings were generated with CLIP (ViT-B/32) [48]. Duplicate posts were then identified using FAISS range search with cosine similarity, applying thresholds of $T_{th} = 0.95$ for text [37] and $I_{th} = 0.99$ for images. Thresholds were chosen to favor high-confidence matches: $T_{th} = 0.95$ follows prior work on near-duplicate text [37], while $I_{th} = 0.99$ sets a stricter bar for images to reduce false positives.

Table 1 summarizes the duplication campaigns uncovered by *OpEx*. Using the above similarity thresholds, *OpEx* identified 76,636 campaigns with 232,581 tweets in US-ELECT24, 210,675 campaigns with 1,020,388 tweets in IND23, and 12,552 campaigns with 40,704 tweets in IND-ELECT24. To validate these results, we randomly sampled 500 text-based and 500 image-based duplicate campaigns. Two authors independently labeled each as either (a) similar or (b) not-similar, and reached full agreement that all sampled campaigns were true duplicates.

This analysis also uncovered several generic, non-narrative campaigns, often triggered by user-installed apps (e.g., Fllwrs, Unfollowspy) that auto-post follower changes or milestones. Other recurring themes included quotes, birthday wishes, holiday greetings, and condolences. Using BERTopic [23], we identified and removed 12,218 generic campaigns from the IND23 dataset and 2,038 from the US-ELECT24 dataset.

Table 2 shows that *OpEx* detects far more duplication campaigns than prior methods. For fair comparison, we re-implemented the baselines of Pacheco et al. [45] and Vishnuprasad et al. [59] on the same account sets and tweet pools as *OpEx*, applying identical pre-processing (removing links, mentions, and hashtags). We also applied the above filtering of generic campaigns across all methods. In US24, *OpEx* (text+image) detected 74,598 campaigns, i.e., 31.5×

Table 3: 10-fold cross-validation on PolitiX (PX) and AltX (AX) shows PAN outperforms Cardaioli et al [15].

	Classifier/Method	Precision (%)	Recall (%)	F1-Score (%)
PX	PAN w RF	88.93	89.00	88.94
	Cardaioli et al. [15]	64.81	69.30	66.75
AX	PAN w RF	92.13	92.38	92.22
	Cardaioli et al. [15]	84.77	64.73	73.44

more than Pacheco et al. [45] and 18.7× more than Vishnuprasad et al. [59]. In IND23, *OpEx* (text+image) identified 198,457 campaigns, 43.9× more than Pacheco et al. [45] and 10.3× more than Vishnuprasad et al. [59]. To confirm validity, two authors manually reviewed a random sample of 500 *OpEx* campaigns and verified that all were true, non-generic duplicates.

Revealing Coordination with Account Similarity Graphs (RQ2).

To further evaluate the impact of multimodal duplication detection (RQ2), we constructed account similarity graphs (ASGs), where two accounts are connected if they duplicate at least dk of the same posts. Following prior work [45, 59], we set $dk = 2$ for U.S. datasets (PolitiX and US-ELECT24). However, we set $dk = 5$ for Indian datasets (AltX and IND23), to reflect their higher posting volume and ensure that only stronger coordination links are captured.

Table 2 shows that in US-ELECT24, *OpEx*’s text-only duplication yielded 3,762 nodes and 28,236 edges, which rose to 4,568 nodes and 41,930 edges once images were included. In IND23, text-only duplication produced 1,724 nodes and 15,345 edges, expanding to 2,077 nodes and 30,047 edges with images. Compared to the best prior work [59], *OpEx*’s ASGs contain up to 4.8× more accounts and 32× more ties (U.S.), and 1.6× more accounts and 4.3× more ties (India), despite stricter thresholds. *OpEx*’s denser ASGs reveal coordination structures that prior methods missed, strengthening transparency and accountability in IO research.

6.3 Attribution (*OpAt*) Evaluation

We evaluated ForeignScope by tuning the β parameter of its label-propagation process. Edge weights in the ASG were normalized, and 10% of seed nodes were held out with priors set to 1/2. Testing β values from 0.1 to 20, we selected $\beta = 10$, which minimized cross-entropy between predicted and known labels. This setup provided stable and reproducible detection of foreign-linked accounts. Substantive results are presented in Section 7.2.

PAN. To evaluate political alignment inference, we built labeled account sets for U.S. (PolitiX) and Indian (AltX) data by combining the NivaDuck dataset [46] with new manual annotations. Labels were R (Republican), D (Democrat), Mixed (M), and Non-political (NP) for PolitiX, and NDA (BJP coalition), INDIA (INC coalition), Other, and NP for AltX. NivaDuck contributed 27 R and 15 D accounts for PolitiX, and 179 BJP, 134 INC, and 23 Other for AltX. Two experts annotated an additional 700 PolitiX and 300 AltX accounts, achieving substantial agreement (Cohen’s $\kappa = 0.78$). After adjudication, the evaluation sets contained 725 PolitiX accounts (296 R, 215 D, 128 M, 103 NP) and 720 AltX accounts (383 NDA, 212 INDIA, 125 NP). Minority classes were oversampled during training only.

Table 3 reports stratified 10-fold cross-validation comparing PAN with Cardaioli et al. [15], using identical train/test splits for fairness. With Random Forest, PAN achieved F1-scores of 88.9% on

Table 4: Identified organization groups with at least 5 accounts in the U.S. (PolitiX) and India (AltX). The observed alignment (θ_o , 3rd column) is at least 60%.

	GID	# Accounts	Observed θ_o	Political Alignment	# Foreign Accounts	Controlled-Campaigns $\theta_c = 0.5$	$\theta_c = 0.8$
US-ELECT24	RG1	796	80.52	Republican	8	6241	5042
	RG2	507	82.2	Republican	50	6728	5964
	RG3	426	84.5	Republican	1	9050	7768
	RG4	394	68.4	Republican	3	5207	4252
	DG1	288	69.4	Democratic	1	6474	5349
	RG5	282	79.5	Republican	2	3466	2832
	RG6	165	55.1	Republican	0	1458	927
	NG1	111	78.3	Non-Political	1	957	884
	DG2	69	81.1	Democratic	0	203	185
	RG7	36	88.8	Republican	14	1540	1472
	RG8	20	95	Republican	0	20	16
IND23	RG9	16	80	Republican	0	686	616
	DG3	11	90.9	Democratic	0	1026	1021
	RG10	8	100	Republican	0	833	803
	RG11	5	80	Republican	0	24	21
	BG1	670	99.85	NDA/BJP	0	37993	31431
	IG1	279	81	INDIA/INC	0	11514	9792
	BG2	176	98.86	NDA/BJP	0	6849	5812
	BG3	168	98.80	NDA/BJP	0	9925	9041
	IG2	166	98.79	INDIA/INC	0	10570	9986
	BG4	73	98.63	NDA/BJP	0	2497	2173
	BG5	39	100	NDA/BJP	0	830	663
	IG3	15	93.33	INDIA/INC	0	569	510
IND23	BG6	12	100	NDA/BJP	0	243	200
	BG7	9	88.88	NDA/BJP	0	109	104
	IG4	6	83.33	INDIA/INC	0	1304	1219
	BG8	5	100	NDA/BJP	0	140	137
	IG5	5	100	INDIA/INC	0	37	37
	BG9	5	60	NDA/BJP	0	207	200

PolitiX and 92.2% on AltX, improving by 22 and 18 points, respectively. These gains reflect PAN’s integration of content, profile, and network features, which improve robustness and generalizability beyond prior models that suffered from limited transferability [17].

7 Findings

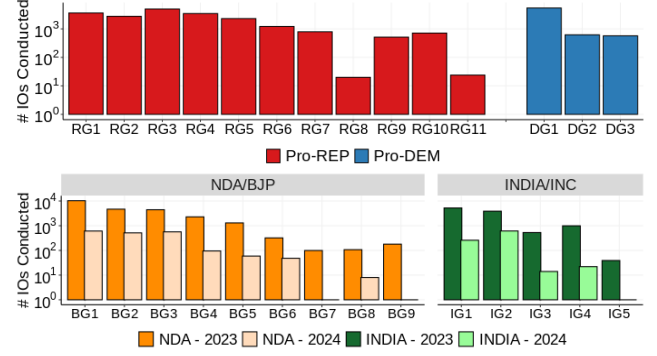
The following analyses address RQ3 by revealing structural and behavioral insights enabled by transparent, large-scale IO detection.

7.1 Structural Insights from Detected IOs (RQ3)

Revealing Operative Groups. From the ASGs identified by *OpEx* (§6), the Louvain community detection algorithm [11] identified 15 account groups in the PolitiX (3,134 accounts) and 14 groups in the AltX (1,628 accounts), each with at least five members (Table 4).

Tracing Campaigns Back to Coordinated Groups. To analyze the structure of coordinated groups (RQ3), we attributed *OpEx*-detected campaigns to the account groups identified above. In US-ELECT24, the above 15 PolitiX groups participated in 63,118 of the above 74,598 campaigns; the above 14 AltX groups participated in 117,547 campaigns. Using a threshold of $\theta_c = 0.5$ (a group must contribute at least half the posts), the PolitiX groups were majority contributors in 43,913 campaigns and the AltX groups in 82,798 campaigns (Table 4). Higher thresholds excluded 11,482 campaigns (avg. size 4.2 posts) with multi-group involvement.

Partisan Attribution of Coordinated Groups. Among the PolitiX (U.S.) accounts, PAN identified 2.25× more Republican than Democrat-supporting users (6,488 vs. 2,878). Among the AltX (India) accounts, PAN labeled 3.8× more NDA/BJP than INDIA/INC-supporting users (3,761 vs. 987). Among the 15 PolitiX groups, 11 leaned Republican, three Democratic, and one non-political, with 11

**Figure 3: Number of IOs detected by IO-X from 14 political groups with at least five members in PolitiX (top) and AltX (bottom). The y axis is log-scaled.**

showing over 75% partisan alignment. Of the 14 AltX groups, nine were NDA-dominated and five INDIA-aligned, with 13 exceeding 80% homogeneity (nine above 95%). Minority-affiliated accounts rarely surpassed 5% per group, likely due to PAN’s false negatives. These patterns reveal how partisanship shapes coordinated IOs.

Uncovering Topics, Narratives, and IOs at Scale. For topic modeling, *OpEx* uses BERTopic [23] and the offline Mistral model [30] to generate narratives. *OpEx* uncovered 64,335 IOs: 27,171 IOs in US-ELECT24 (from 10,668 topics and 16,879 narratives), 34,360 IOs in IND23 (12,626 topics, 18,763 narratives), and 2,804 IOs in IND-ELECT24 (1,050 topics, 1,662 narratives). These IOs are the basis form our partisan and harmful-content analyses in Section 7.2.

7.2 Partisan Asymmetries in IOs (RQ3)

From US-ELECT24 and PolitiX, *OpEx* extracted 27,171 IOs with 10,651 topics: 20,523 from Republican and 6,648 from Democratic-supporting groups. In IND23 and AltX, *OpEx* identified 34,360 IOs with 12,625 topics, including 23,685 by BJP and 10,675 by INC-supporting groups. In IND-ELECT24, IO-X identified 1,897 BJP and 907 INC-led IOs. The IO counts exceed narratives because multiple groups often promote the same narrative.

Manual review of 100 random topics (50 PolitiX, 50 AltX) showed alignment with the political affiliations of promoting groups and divergence across opposing sides. A second 100-topic review confirmed consistency among groups backing the same organization.

Figure 3 shows the number of IOs by each operative group in PolitiX and AltX. In PolitiX, the most active R-supporting group ran 5,017 IOs, while the top D-supporting group ran 5,450. Although more groups leaned Republican, one Democratic group (DG1) was the most active. In AltX, the top NDA/BJP group ran 10,287 IOs, almost double the 5,241 by the leading INDIA/INC group. Five groups each exceeded 100 IOs; the top NDA/BJP and INDIA/INC groups almost tied (609 vs. 612). These results reveal IO-X’s ability to expose partisan asymmetries in coordinated influence operations.

7.3 Cross-Border Participation (RQ3)

Applying ForeignScope revealed 965 foreign-linked accounts in PolitiX, defined as those posting in foreign languages or inferred

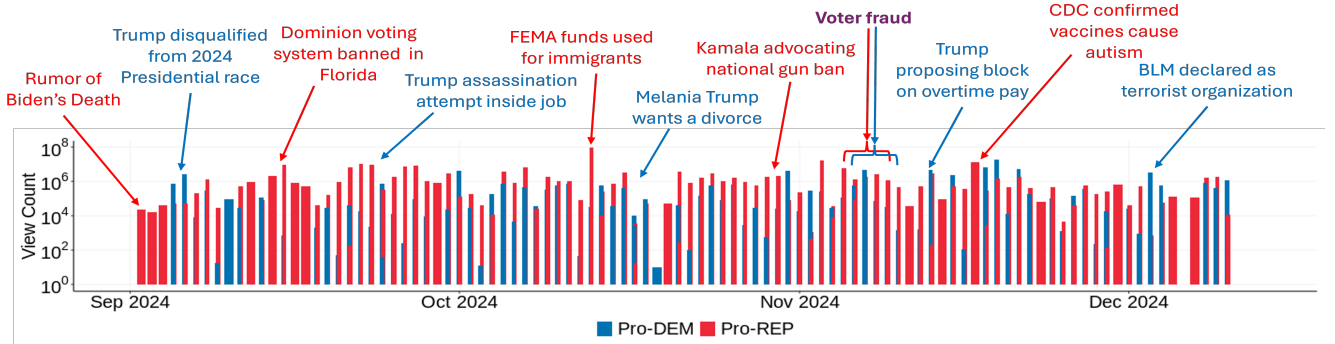


Figure 4: Timeline of 957 IO-X-identified campaigns promoting false information: 691 by Republican-supporting groups, 266 by Democratic-supporting groups. Y-axis shows total views. Republican campaigns received 3.9x more views (324M vs. 83M).

via label propagation, with activity patterns aligned to corresponding time zones. These included Latin-American (684), Indian (135), Brazilian (67), Chinese (20), Israeli (19), Persian (15), Russian (14), and Ukrainian (11). PAN identified a $2.7\times$ imbalance (239 R vs. 87 D) among the 965 foreign-linked accounts, concentrated in Latin America, India, Israel, and Russia. Several operative groups contained foreign members (Table 4), such as RG2 (10% foreign, including India, China, Israel, and Russia) and RG7 (14 foreign accounts, primarily Latin American and Indian). In contrast, AltX contained only 37 foreign-linked accounts, mostly from Arabic countries (21) and Iran (11), with none embedded in operative groups.

This asymmetry reveals geopolitical differences in IO strategies: Indian campaigns were overwhelmingly domestic, while U.S. operations exhibited visible cross-border participation.

7.4 False and Harmful Campaigns (RQ3)

IO-X Campaigns Referencing Untrustworthy Sites. We compiled two datasets of untrustworthy domains (5,720 sites total) from fake-news lists [3] and specious sources [24]. In US-ELECT24, IO-X detected 14,463 campaigns using such URLs (8,469 tied to IOs), with Republican groups running $7.7\times$ more than Democratic groups (7,522 vs. 974). In IND23 and IND-ELECT24, IO-X found 1,432 such campaigns, almost all (1,236) run by NDA/BJP groups. Thus, untrustworthy sources are disproportionately concentrated in Republican and BJP-led IOs.

False Information Campaigns. Claimbuster [25] flagged 1,474 false campaigns in US-ELECT24; 957 of these were part of IO-X-detected IOs (691 R vs. 266 D). Figure 4 shows their timeline, with R-groups drawing 324M views compared to 83M for D-groups. In India, we identified 183 false campaigns from AltNews24; manual annotation of 500 additional cases confirmed 61 (IND23) and 26 (IND-ELECT24) as false.

These campaigns averaged 23 duplicates and reached millions: in the U.S., 1,327 campaigns exceeded 1M views, versus 67 in India (max 4.25M). Higher engagement in the U.S. reflects larger datasets, algorithmic amplification, and more influential accounts.

Hate and Offensive Campaigns. TweetNLP [14] found 1,105 hate and 4,642 offensive campaigns in US-ELECT24, of which 811 hate (740 R, 71 D) and 3,378 offensive (2,556 R, 822 D) tied to IOs. In IND23 and IND-ELECT24, 2,401 hate and 5,866 offensive campaigns

were detected. Manual checks showed these often intertwined with false claims. Overall, falsehoods and toxicity frequently co-occurred, often amplified by verified or influential users.

This analysis reveals IO-X’s ability to uncover the scale and partisan asymmetries of harmful narratives, enabling transparent oversight of coordinated misinformation.

8 Discussion

IO-X faces several constraints. Attribution relies on heuristic signals and threshold choices (e.g., in PAN and ForeignScope), which adversaries could attempt to game. Validation is limited to annotated subsets. Without full ground truth it remains difficult to separate coordinated inauthenticity from organic repetition. Our analysis also depends on fact-checker reports and partial platform data, leaving gaps in some campaign timelines. These limitations reveal opportunities for future work, including engagement-weighted sentiment modeling, adaptive thresholds, larger annotated datasets, and more comprehensive data access.

Despite these challenges, IO-X successfully uncovered thousands of accounts, including hundreds of foreign-linked actors, involved in tens of thousands of content duplication campaigns and influence operations during the 2024 elections in both the U.S. and India. This shows that meaningful insights can be achieved using only public data and transparent methods. Importantly, IO-X enables reproducible, accountable oversight of manipulative online behavior despite adversarial pressures and platform opacity.

9 Conclusions

We introduced IO-X, a system for detecting content-duplicating influence operations using multimodal duplication, foreign-link analysis, and partisan attribution. Applied to fact-checker-seeded data from the 2024 U.S. and Indian elections, IO-X uncovered tens of thousands of coordinated campaigns, showing that transparent and reproducible detection can advance accountability on the web.

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Appendix

A Proof of Claim 1

PROOF. We prove the contrapositive: if any one of *Mon*, *OpEx*, *OpAt* is missing, an IODetSys cannot attain non-negligible advantage for *attributed* IO detection in the IO-UND game.

(No Mon). Without monitoring, the detector observes no new posts or accounts across rounds; its knowledge state K is invariant in $i = 1, \dots, r$. Hence the transcript seen by the detector is independent of the hidden bit b , and any decision rule has $\Pr[b' = 1 \mid b=1] = \Pr[b' = 1 \mid b=0]$, implying $\text{Adv}(\mathcal{A}) = 0$.

(No OpEx). Without operation extraction, the detector cannot compute cross-account, cross-round coordination statistics (e.g., intersections across the r nominated topic–narrative pairs). Its decision can only depend on per-round marginals that are indistinguishable between coordinated and background behavior once account identities are not aggregated across rounds. Thus the distribution of any statistic it can compute is the same under $b=1$ and $b=0$, yielding $\text{Adv}(\mathcal{A}) = 0$.

(No OpAt). For *attributed* detection, success requires both detecting coordination and assigning the responsible organization in $\mathcal{O}$. Without attribution, the system cannot produce a label in $\mathcal{O}$; the probability of “detect and correctly attribute” is therefore 0 (or, at best, baseline chance over labels), which is negligible. Hence the end-to-end advantage for attributed IO detection is negligible.

Since in each case the advantage collapses when one capability is absent, the three capabilities are jointly necessary for non-negligible advantage. $\square$

B IO-X Pseudocode

Algorithm 1 Data collection process. The ReportCollector thread collects fact-checked reports once per ST interval. The MonitorAccounts thread collects Twitter/X data from monitored accounts who duplicated reported false claims.

Require: FactChecker ▷ Fact-checking site
Require: priorityQ = $\emptyset$ ▷ Queue of monitored accounts
Require: duplicatorPosts = $\emptyset$ ▷ Set of posts of monitored accounts
Require: reports ▷ Latest PolitiFact reports

```

1: procedure REPORTCOLLECTOR(int ST)
2:   while true do
3:     reports = FactChecker.getLatestReports()
4:     for  $r \in$  reports do
5:       tweets = r.getTweets()
6:       if tweets  $\neq \emptyset$  then
7:         for  $t \in$  tweets do
8:           tweets.add(X.search(t))
9:       else
10:        tweets.add(X.search(r.reportTitle))
11:      for  $t \in$  tweets do
12:        account = getPosterAccount(t)
13:        priority = account.assignInitPriority()
14:        priorityQ.add(account, priority)
15:      sleep(ST)
16: procedure MONITORACCOUNTS
17:   while priorityQ  $\neq \emptyset$  do
18:     account = priorityQ.getFirst()
19:     if (X.available(account) == false) then
20:       priorityQ.remove(account)
21:     else
22:       posts = X.getLatestPosts(account)
23:       duplicatorPosts.add(posts)
24:       priorityQ.add(account, updatePriority(account, posts))

```

Algorithm 1 shows the pseudocode for the monitoring function *Mon* of IO-X. Data collection runs in two parallel processes. The ReportCollector periodically fetches new fact-checking reports (line 3), extracts referenced tweets (line 5), and searches for duplicates or title-matching tweets depending on report content (lines 6–10). For each tweet found, it retrieves the posting account (line 12), assigns a default priority based on the estimated time to produce k posts (line 13), and queues the account (line 14).

The MonitorAccounts process tracks activity of accounts in the priority queue. It removes the highest-priority account (line 18), discards it if inaccessible (lines 19–20), otherwise collects its latest unique posts, i.e., removing self-duplicated posts (line 22), updates its priority based on post activity (line 23), and reinserts it into the queue (line 24). Accounts posting more frequently get higher priority, and ties are broken by the last collection time. This ensures more frequent post collection for more active accounts.